

Defense Preparation Dossier —

Arguente: Prof. Paulo Jorge Azevedo

Who he is: Assistant professor, Dep. Informática, U. Minho; senior researcher at INESC TEC (HASLab). PhD from Imperial College (logic programming). **He is the author of the CAREN engine and co-author of the Distribution Rules paper your entire thesis is built on.** Research: association rules, distribution rules, subgroup discovery, rule interest measures, time-series motifs, model-performance analysis via rules.

His publications most relevant to you:

Paper	Why it matters for your defense
Jorge, Azevedo & Pereira (2006), <i>Distribution Rules with Numeric Attributes of Interest</i> , PKDD 2006, LNAI 4213, pp. 247–258	The DR formalism + KS test you use. He co-invented it. Know it cold.
Azevedo (2003), <i>CAREN — a Java based Apriori implementation for classification purposes</i> , Tech. Report, U. Minho	CAREN is his engine — Apriori-based, exhaustive levelwise search.
Azevedo & Jorge (2007), <i>Comparing Rule Measures for Predictive Association Rules</i> , ECML	His work on interest measures (conviction wins). He may ask why you weight rules by support rather than by an interest measure.
Pimentel, Azevedo & Torgo (2023), <i>Subgroup mining for performance analysis of regression models</i> (Error Distribution Rules), Expert Systems	He uses rules to analyze where regression models fail — exactly what your era-composition analysis does informally. Expect a question; cite it.
Soares, Azevedo, Cerqueira & Torgo (2025), <i>Meta Subspace Analysis</i> , Discovery Science 2025	Recent extension of the same idea to metafeature space.

Strategic framing: he will be *sympathetic* to the method (it's his) and *demanding* on how faithfully you describe it and how rigorously you evaluate it. Your risk is not "why CarenR?" — it's misdescribing CAREN and sloppy statistics.

////////////////////////////////////

PART 0 — CRITICAL: HE IS READING THE SUBMITTED (UNCORRECTED) VERSION

The jury reads the version you deposited — **the errors are still in his copy**. You cannot rely on the fixes being visible to him; you must be able to answer *out loud* when he points at the printed page. The universal move:

Acknowledge instantly → **state the correct fact** → **note it is on your errata sheet**. No over-apologising, no defensiveness. He is meticulous about his own work; owning an error cleanly *raises* your credibility, flailing lowers it.

Bring the printed **errata sheet** and offer it at the start ("I identified some editorial corrections after submission and prepared an errata; may I hand it to the jury?"). Ask your supervisor if handing it up front is the local custom — if yes, it pre-empts half of these.

If he points to the submitted page, say this:

He points at (his copy)	What he sees	Your spoken answer
p.36 — "CAREN ... originally developed by Jorge et al. (2006)"	CAREN credited to the 2006 paper	"That's a mis-attribution I've corrected — the CAREN engine is your 2003 work; Jorge, Azevedo and Pereira 2006 added the distribution-rule layer on top. It's item 1 on my errata." <i>(Then state the provenance confidently — it flatters him and proves you know the</i>

		<i>lineage.)</i>
p.36 / p.28 — "beam search" / "breadth-first, most promising conditions"	Two different, both wrong, search descriptions	"CAREN does an Apriori-style levelwise search pruned by minimum support — exhaustive over the support-frequent itemsets, not a heuristic beam. The two passages were inconsistent; corrected to that. I can sketch the lattice if useful." (<i>Be ready to draw it — see Part 1 item 4.</i>)
p.36–38 / Appx A, B — "support $\geq 15\%$ "	15% support threshold	"The value actually used in the code is 20% — <code>carenr_min_sup = 0.2</code> . The results chapters are already at 20%; the earlier chapters lagged. My Appendix E logged the correction; the errata propagates it. So: a rule must cover ≥ 18 of 89 years in RDD."
p.99 — DOI <code>11871637_25</code>	Wrong DOI on the 2006 paper	"Typo — correct DOI ends <code>_26</code> , LNCS 4213, pp. 247–258. On the errata."
p.100 — "Contrast		"That title expansion was wrong and the TODO shouldn't have printed — both removed.

Association Rules for Extraordinary Nodes" + "TODO: confirm..."	Fabricated title + a literal TODO in the bibliography	I'm confirming the exact package citation with Prof. Moreira." (<i>Don't invent a replacement title on the spot.</i>)
p.99–100 — "TODO: confirm..." on Fraga/Lopes	Placeholder notes visible in refs	"Editorial placeholders that slipped through; removed, metadata being confirmed with my supervisors."

Tone rule: each answer is one or two sentences. Acknowledge, correct, move on — do **not** spiral into apology, and do **not** argue the error doesn't matter. "Good catch, here's the correct version, it's on my errata" is the whole play. None of these touches a result, and you can say that plainly if he asks the significance: *"none of the corrections affects any table, test, or conclusion — the 20% threshold was used in all computations; only the prose lagged."*

PART 1 — CORRECTIONS (already applied in your working LaTeX; his copy still has them)

These were factual errors about *his own work*. **All are now fixed in the LaTeX and the PDF recompiles cleanly** (see `ERRATA.md`). Listed here so you know what changed and never say the old versions aloud. If any is raised in the defense, the honest answer is "yes — that's on my errata sheet, already corrected."

1. **CAREN mis-attribution [FIXED]**. Ch3 said CAREN was "originally developed by Jorge et al. (2006)". Wrong: the CAREN engine is **Azevedo's** (2003 tech report, U. Minho); Jorge, Azevedo & Pereira (2006) added the **Distribution Rules** framework on top. Now reads "...developed by Azevedo (2003) and extended to distribution rules by Jorge et al. (2006)". New `Azevedo2003` bib entry added. **Say it this way out loud when you introduce the method.**
2. **Wrong DOI [FIXED]**. Was `10.1007/11871637_25` ; corrected to

10.1007/11871637_26 (PKDD 2006, LNCS 4213, pp. 247–258, Springer), with volume/pages added.

3. **Invented backronym [FIXED].** "CarenR: Contrast Association Rules for Extraordinary Nodes" was a fabricated expansion carrying a printed TODO. Removed; the entry now neutrally describes the R package. **Still confirm the exact Moreira (2022) citation with Prof. Moreira** — the definitive version should not guess. (CAREN itself traces to Azevedo's "class project association rule engine".)
4. **Search-strategy inconsistency [FIXED].** Ch2 said "beam search", Ch3 said "breadth-first, most promising conditions" — contradictory, and neither matched CAREN, which is **Apriori-style levelwise search pruned by minimum support**. Both now say that (verified against the `carenr` package call in `6_scenario_validation.R`). This was the single most likely whiteboard question — **be ready to draw the levelwise lattice**: level 1 = all single feature-bin conditions passing min-support; level k built from support-frequent level- $(k-1)$ itemsets; KS test applied to each surviving subgroup; max length 3–4.
5. **Support threshold [FIXED — value is 20%].** Text said 15% in Ch2/Ch3/Appx A/B but 20% in Ch4–6. The **code uses 0.20** (`carenr_min_sup <- 0.2`), so 20% is correct; all text now says 20% (and derived counts: ≈ 18 years of 89). If asked "what was your support threshold?", the answer is **20%, i.e. a rule must cover ≥ 18 of 89 years (RDD)**. Note this correction was already logged in your Appendix E — it had just not been propagated to every chapter.
6. **Printed TODO notes [FIXED].** The submitted bibliography printed literal "TODO: confirm..." notes on the Moreira, Fraga and Lopes entries — visible to any reader. Removed. **Confirm the real metadata for all three with your supervisors** before the definitive submission.
7. **Terminology.** In his vocabulary: *distribution rules (DR)*, *antecedent/consequent*, *support*, *KS interest*. Don't say "beam", "gain", or "coverage heuristic" for CAREN.

Turn Appendix E into an asset. Your AI-transparency appendix already documents a numerical self-verification pass that *caught* the 15→20% error (among others). If the support-threshold or any numeric point comes up, note that the thesis includes an explicit verification appendix — it demonstrates exactly the critical-engagement rigor an examiner wants to see, and reframes "there were errors" as "I built a process that finds

and fixes them."

PART 2 — LIKELY QUESTIONS AND PREPARED ANSWERS

A. CAREN / Distribution Rules internals (his home turf)

Q1. "Explain exactly what a distribution rule is and how your usage differs from the 2006 paper." A: A DR is IF antecedent THEN empirical distribution of the numeric property of interest, deemed interesting when the KS test rejects equality with the default (whole-dataset) distribution. My usage follows the 2006 formalism; the thesis-level additions (explicitly separated in Fig. 3.x) are: the property of interest is a *detrended residual* computed within-fold; a support filter of 20%; $p \leq 0.10$; support-weighted aggregation of subgroup medians for point prediction; and a fallback to the training median when no rule fires. The algorithm is unchanged — the contribution is the temporal validation harness around it.

Q2. "You test each subgroup against the full dataset, which contains the subgroup itself. The two samples are not independent." A: Correct — this follows the original DR design (subgroup vs. default distribution), which makes the test conservative in the direction that matters: overlap pulls the two ECDFs together, so D is understated, not overstated. Comparing subgroup vs. complement is a known alternative; with $n=80-89$ and 20% support the difference is modest, and I kept the default-distribution convention for consistency with the DR literature.

Q3. "Thousands of candidate subgroups, $p \leq 0.10$, no multiple-testing correction. How many of your 48 RDD rules are false discoveries?" This is the single most probable hard question. A: I don't rely on the raw p-values alone for validity. Four independent guards: (1) 20% minimum support (≥ 18 of 89 years) removes the most fragile candidates; (2) the LOESS re-validation shows the top features retain 90–92% of their correlation once era co-trending is removed — an era-artefact rule would not survive that; (3) RIPPER and M5Rules converge on the same variable families under completely different criteria; (4) the walk-forward evaluation is itself an out-of-sample test of the rules. I present the rule set as validated *exploratory* findings, not confirmatory hypotheses — and the honest statistical framing in Ch5 (confirmatory vs. exploratory tests, uncorrected p-values disclosed) applies the same standard. If pressed:

acknowledge a permutation/FDR analysis (e.g., swap-randomization of the target) is a natural extension.

Q4. "Why $p \leq 0.10$ and not 0.05?" A: A deliberate sensitivity/power trade-off at $n=80-89$: with 20% support a subgroup has $\sim 18-25$ observations, where the KS test still has limited power; 0.05 empties the rule set in several folds. Since rules face a second, harsher filter — out-of-sample MAE — the lenient discovery threshold is compensated downstream. (Know the number: how many rules survive at 0.05. If you don't, run it before the defense.)

Q5. "The KS test is asymptotic; with a subgroup of 12 years, are the p-values even valid? And ties?" A: Subgroup sizes are small but the reference set is 60–89; the exact/asymptotic distinction at these sizes shifts p-values slightly, which is one more reason the rules are treated as exploratory. Residuals are continuous, so ties are rare. (If you used R's `ks.test`, it computes exact p-values for small samples without ties — check which the *caren* package uses and say so.)

Q6. "The DR paper filters redundant rules against their ancestors (improvement). Do you?" A: Yes — Section 4.x analyses rule overlap and unique-vs-redundant rules; the rule sets organise around a small number of backbone features with contextual refinements. (Re-read §4 "Rule Set Overlap and Internal Coherence" the week before — be able to give the counts.)

Q7. "Why quartile equal-frequency discretisation of antecedents? CAREN supports supervised discretisation (Fayyad–Irani)." A: I tested exactly that — *CarenR_Supervised* derives cut-points maximising residual separation within each training fold. It was the **worst** variant in both regions (240/347 mhl): the bin boundaries are optimised on the same data that assesses rule quality, a target-leakage effect that inflates rule counts (46–146 per scenario) and collapses out of sample. Equal-frequency quartiles guarantee uniform bin occupancy, which the KS test needs at this n . This negative result is one of the replicable findings of the thesis.

Q8. "You aggregate fired rules by support-weighted medians. Why support and not an interest measure — KS statistic, conviction?" A: I evaluated confidence-style weighting (*CarenR_Dist_Conf*, weights from KS p-values) and stacking; the aggregation mechanism turned out to be second-order — variant ranking is driven by feature selection, not weighting (§3.x). Support weighting favours estimates whose subgroup medians are computed on more observations, i.e., reliability weighting. If he presses on conviction: conviction is defined for categorical consequents; for DRs the natural

analogues are KS-derived, which is what Dist_Conf tested.

Q9. "DRs give you an entire distribution. Why throw it away and predict only the median? Why not evaluate distributional forecasts (CRPS, quantile loss)?" A: Fair criticism and a very natural extension. The point forecast was chosen for comparability with the baselines and the agronomic use case (a single expected-production figure). But the fallback design already exposes the distributional view — when no rule fires the forecast *is* the training distribution's median. Evaluating the full predictive ECDF with CRPS is listed as future work; with 17 test years per region the CRPS estimates would themselves be noisy. (If this isn't in future work yet — add one sentence to ch7.)

B. Evaluation methodology (his EDR / model-performance angle)

Q10. "Your Wilcoxon test treats 18 expanding-window scenarios as independent samples. They share most of their training data and their test sets overlap." A: Correct, the scenarios are positively correlated, which inflates the effective sample size of any test. This affects both directions equally and mostly makes the tests *less* likely to reach significance — consistent with the honest null result ($p=0.468$). The RVV negative result ($W=0$, $p=0.004$) is a unanimous sign pattern across all 9 rule-firing scenarios, robust to the dependence caveat. I flag the p-values as indicative, not exact.

Q11. "You selected the best of 11 CarenR variants after seeing the results, then compared it to the baseline." A: Yes — and that selection bias *strengthens* the headline conclusion, because even the post-hoc best variant fails to beat Naive_Median (3% behind in RDD, 22% in RVV). The bias would only matter if I claimed victory. The one positive claim (era-balanced scenarios 10–14, $p=0.031$) is explicitly labelled post-hoc/exploratory throughout, requiring prospective validation.

Q12. "Your era-composition analysis asks 'where does the model fail'. Are you aware this is exactly what Error Distribution Rules do?" A: Prepare to say YES. "Pimentel, Azevedo & Torgo (2023) formalise this as subgroup mining over model errors — EDRs. My era-composition analysis does this manually with one covariate (post-1990 training share). Applying EDRs to the walk-forward error series, with era share and climate features as descriptors, would be the principled generalisation and is an excellent direction for the future-work section." Consider actually adding a citation to this paper in ch6/ch7 before the defense — it is respectful, correct, and defuses the question.

Q13. "Why weighted MAE? Weighting by test-set size makes early, era-confounded scenarios dominate." A: That interaction is analysed explicitly in §6.4: weighting by test size is the statistically correct choice (more observations, more influence), and I report per-scenario MAE and mean MAE alongside so the reader sees the era structure. The sensitivity is a property of the data's era composition, not an artefact I hid.

Q14. "Naive_Median is a straw man / why is it so hard to beat?" A: It's the opposite of a straw man — it's the MAE-optimal constant predictor on the residual scale, so beating it is precisely the claim "climate features carry exploitable signal." The two-decision decomposition separates trend accuracy (Decision 1) from climate signal (Decision 2), so the baseline is doing diagnostic work, not just benchmarking.

C. Results and interpretation

Q15. "Goal 2 failed. What did we learn?" A: Three transferable findings: (1) the discovery–prediction asymmetry is a property of the system — $|r| \approx 0.47 \Rightarrow \sim 22\%$ variance explained is enough for statistically valid pattern discovery, not for beating a median at $CV \approx 47\%$ noise; (2) the two-decision decomposition localises the bottleneck in structural detrending, not rule learning — time as sole detrending covariate conflates industry restructuring with climate; (3) confirmed negative results with practical value: supervised discretisation leaks, LOESS consumes the signal it should expose, and RVV rules actively harm ($p=0.004$). The path forward is structural covariates (area, beneficio quota, training-system share) in the detrend model — data exists at IVV/IVDP.

Q16. "In RVV your rules make predictions worse. Shouldn't the method detect that its own rules are era artefacts?" A: That's the correct reading — and it's why the RVV result matters. The KS test validates rules against the *training* distribution; when train and test straddle a structural break (1986 EU accession, pergola→espalier conversion, –65% production), statistically valid training-era rules encode era membership. CarenR partially resists this (the detrending reframing, §6.6), but no within-sample statistic can certify out-of-era generalisation. That's an argument for the walk-forward protocol itself — it caught what discovery-only evaluation would have missed.

Q17. "Only 17 test years per region. Are any of these conclusions stable?" A: Acknowledged explicitly in ch2 (§ limitations): 17 years spans ~ 3 cycles of the dominant 5.7-year oscillation. This is why the thesis reports per-scenario results, distinguishes confirmatory from exploratory tests, and avoids strong positive claims. The alternative —

random CV — would be temporally dishonest and inflate performance.

Q18. "What exactly is new here versus the group's previous CarenR work (Moreira et al.)?" A: Previous work applied CarenR for discovery on agricultural data. This thesis adds: first walk-forward predictive validation of DRs (fold-internal discretisation, feature selection, and rule induction — zero leakage); the two-decision decomposition as a diagnostic; the era-confound identification and LOESS re-validation methodology; the 11-variant structured sensitivity analysis; and the cross-regional structural comparison. In short: the thesis converts CarenR from a discovery tool into a fully evaluated forecasting pipeline, and shows precisely where and why the forecasting half breaks.

Q19. "Why not Exceptional Model Mining / pysubgroup / Cortana? Why CAREN specifically?" A: DRs are the right primitive: the target is a single numeric variable and the question is distributional shift — exactly the DR interestingness criterion, with the KS test giving nonparametric validity at $n < 90$. EMM would fit a model per subgroup — with ~ 12 – 25 observations per subgroup there is no capacity for that. And CAREN's exhaustive support-pruned search gives completeness guarantees a heuristic searcher doesn't. (Bonus: this answer compliments his design decisions — accurately.)

Q20. Rapid-fire facts to have memorized.

- RDD: $n=89$ (1934–2022), 18 scenarios; RVV: $n=80$ (1942–2021), 16 scenarios; 17 test years each.
- RDD: Naive_Median 184.0 vs CarenR_Dist_Eta2 189.2 (3% gap); RVV: 140.4 vs Sup_FS 171.6 (22%).
- Tests: RDD Wilcoxon $W=68$ $p=0.468$ (confirmatory, ns); RDD era-balanced sign test $p=0.031$ (post-hoc); RVV rule-firing Wilcoxon $W=0$ $p=0.004$ (confirmatory negative).
- Rules: 48 RDD, 20 RVV; support $\geq 20\%$ ($\approx 18/89$ yrs); $p \leq 0.10$; quartile equal-frequency bins; 59 features; max rule length 3–4.
- LOESS span 0.40: RVV baseline $140 \rightarrow \sim 90$ but 0 rules; RDD baseline $184 \rightarrow 290$.
- LOESS retention of top features: 90–92% (harvest temperature only 56% — era-inflated, correctly not selected by CarenR).

PART 3 — DEFENSE STRATEGY

1. Part 1 items are already fixed in the LaTeX (PDF recompiles cleanly). Bring the

errata sheet (`ERRATA.md`) to the defense; if any item comes up, "already on my errata list" turns it into a point in your favour. Still to do on your side: confirm the Moreira/Fraga/Lopes citation metadata with your supervisors for the definitive deposit.

2. **Cite him back.** Work one slide/sentence connecting your era-failure analysis to his EDR paper (2023). Examiners soften when their line of work is engaged seriously, and here the connection is real, not flattery.
3. **Own the negative result.** Your strongest asset is the honest statistical framing (confirmatory vs. exploratory, disclosed uncorrected p-values, leakage-free folds). Lead with it; he values it — his own papers are meticulous about evaluation.
4. **Search strategy — confirmed and fixed.** The `carenR` package call performs CAREN's support-pruned levelwise (Apriori-style) candidate generation; the thesis-level cap is rule length 3–4. Safe verbal answer: "the R package inherits CAREN's support-pruned levelwise candidate generation; I cap rule length at 3–4." Be ready to sketch the lattice (see Part 1, item 4).
5. **Prepare the one number you don't have:** how many rules survive at $p \leq 0.05$, and rule counts in the redundancy analysis. He asks for numbers.

PART 4 — WHITEBOARD: "How does CAREN search?" (the most likely live question)

If he asks how CAREN generates rules, **draw a three-row lattice and talk through it**. This is his algorithm — getting it right here is worth more than any slide.

What an "item" is

An *item* is a single feature-bin condition, where the bin is an **explicit numeric interval** produced by your Step-1 equal-frequency (quartile) discretisation — **not** a categorical label like "low." For example `swa_hv_y1_b20 ∈ [20–74 mm]` (the bottom quartile of pre-harvest soil water), or `∈ [74–101 mm]`, `∈ [101–152 mm]`, `∈ [152–350 mm]`. Each of your 59 features contributes 4 such interval-items, so the pool is 59×4 items. The search explores *conjunctions* of these interval conditions, one level at a time.

(Be precise on this if he probes: bins are quartile intervals with real boundaries computed on the training fold — equal-frequency, so ~22 of 89 obs per bin in RDD — and a rule antecedent is a conjunction like `swa ∈ [20–74] ∧ rain_f1 ∈ [...]` .

Say "interval," never "low/high," when describing an item.)

Draw three rows

Label the nodes with **interval items** (write the interval, not "low/high"). Use short names

— `swa` = pre-harvest soil water, `rain_fl` = wet days post-flowering, `heat_sm` = hot days pre-maturity, `tmax_hv` = max temp at harvest:

- **Level 1 — single interval conditions.** Compute each condition's **support** (fraction of years it covers). Keep those $\geq 20\%$, cross out the rest. (In the sketch: `tmax_hv` $\in [28, 31]$ dropped at 12%.)
- **Level 2 — pairs, formed only from level-1 survivors.** Compute support again; drop those below 20%. (`swa` $\in [20, 74]$ $\wedge$ `heat_sm` $\in [3, 9]$ dies at 9%.)
- **Level 3 — triples, formed only from surviving pairs.** The triple is **never generated**, because one of its pairs already failed. Draw it as a ghost box. Cap length at 3–4 conditions.

```

                A = swa $\in[20,74]$     B = rain_fl $\in[8,15]$     C =
heat_sm $\in[3,9]$     D = tmax_hv $\in[28,31]$ 

Level 1:    [A ✓ 34%][ ]    [B ✓ 44%][ ]    [C ✓ 27%][ ]    [D ✗ 12%]
← support filter

                \          /    \          /
Level 2:    [A $\wedge$ B ✓ 22%][ ]    [A $\wedge$ C ✗ 9%][ ]    [B $\wedge$ C ✓ 21%]
← support filter

                \          |          /
Level 3:    ( A $\wedge$ B $\wedge$ C never generated )
pruned via A $\wedge$ C

```

Each node is a **conjunction of intervals**; a surviving node that also passes the KS test becomes a rule like "IF `swa` $\in [20,74]$ mm AND `rain_fl` $\in [8,15]$ days THEN production distribution shifts +... mhl."

The one idea that makes it worth drawing — anti-monotonicity

Support can only **stay equal or shrink** as you add interval conditions (a narrower conjunction covers fewer or equal years). So if a small itemset is already below threshold, every larger itemset containing it is guaranteed to be below too — you skip it without testing. That is the cascade: dropping `tmax_hv` $\in [28, 31]$ removes all its

pairs; dropping $A \wedge C$ removes the $A \wedge B \wedge C$ triple. This is what keeps an exhaustive search tractable.

Two filters, not one (say this explicitly)

Support is only the **first** filter — it *builds the lattice*. Then the **KS test** runs on each surviving subgroup, comparing its residual distribution against the global one; subgroups with $p \leq 0.10$ become distribution rules (the green boxes). So there are two distinct criteria: **support = structural/coverage**; **KS = statistical**. Students routinely conflate them — keeping them separate signals that you understand his method.

Why "exhaustive, not greedy" matters (turns the old error into a strength)

Contrast it aloud with the discarded "beam search" wording. A beam/greedy learner keeps only the top-*k* branches per level and can permanently miss a good combination. CAREN keeps **all** frequent candidates, so it has a **completeness guarantee** over the frequent region — a genuine strength of his design.

The sentence that answers the whole question

"Levelwise candidate generation with Apriori support-pruning by anti-monotonicity — exhaustive over the frequent itemsets, not a heuristic beam — and then the KS test as a second, statistical filter that selects the rules. I cap rule length at 3–4 conditions."

Caveat — stay honest: the supports in the sketch (34%, 12%, 9%...) are illustrative, not from your data. Draw the *structure*; don't quote those specific numbers as real results.